

UNIT – V**FILES, EXCEPTIONS, MODULES, PACKAGES**

Files and exception: text files, reading and writing files, command line arguments, errors and exceptions, handling exceptions, modules (datetime, time, OS , calendar, math module), Explore packages.

Files, Exceptions, Modules, Packages:**Files and exception:**

A **file** is some information or data which stays in the computer storage devices. Python gives you easy ways to manipulate these files. Generally files divide in two categories, text file and binary file. Text files are simple text where as the binary files contain binary data which is only readable by computer.

- **Text files:** In this type of file, Each line of text is terminated with a special character called EOL (End of Line), which is the new line character ('\n') in python by default.
- **Binary files:** In this type of file, there is no terminator for a line and the data is stored after converting it into machine understandable binary language.

An **exception** is an event, which occurs during the execution of a program that disrupts the normal flow of the program's instructions. In general, when a Python script encounters a situation that it cannot cope with, it raises an exception. An exception is a Python object that represents an error.

Text files:

We can create the text files by using the syntax:

Variable name=open (“file.txt”, file mode)

For ex: f= open ("hello.txt","w+")

- We declared the variable f to open a file named hello.txt. **Open** takes 2 arguments, the file that we want to open and a string that represents the kinds of permission or operation we want to do on the file
- Here we used "w" letter in our argument, which indicates write and the plus sign that means it will create a file if it does not exist in library

- The available option beside "w" are "r" for read and "a" for append and plus sign means if it is not there then create it

File Modes in Python:

Mode	Description
'r'	This is the default mode. It Opens file for reading.
'w'	This Mode Opens file for writing. If file does not exist, it creates a new file. If file exists it truncates the file.
'x'	Creates a new file. If file already exists, the operation fails.
'a'	Open file in append mode. If file does not exist, it creates a new file.
't'	This is the default mode. It opens in text mode.
'b'	This opens in binary mode.
'+'	This will open a file for reading and writing (updating)

Reading and Writing files:

The following image shows how to create and open a text file in notepad from command prompt

```
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

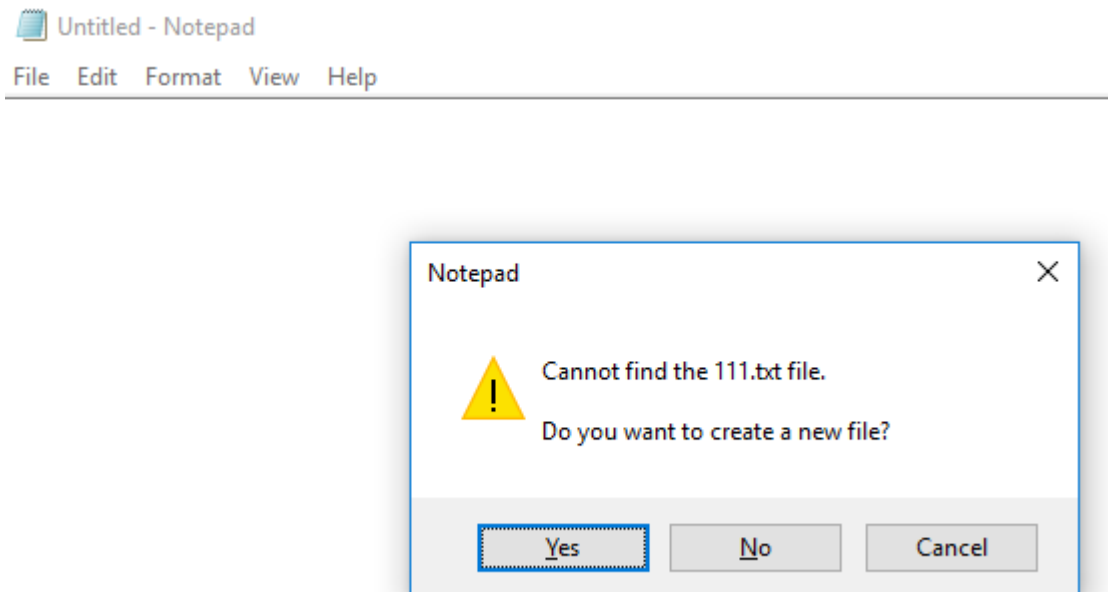
C:\Users\MRCET\AppData\Local\Programs\Python\Python38-32\filess>start notepad hello.txt

C:\Users\MRCET\AppData\Local\Programs\Python\Python38-32\filess>type hello.txt
Hello mrcet
good morning
how r u
C:\Users\MRCET\AppData\Local\Programs\Python\Python38-32\filess>
```

(or)

```
C:\Users\MRCET\AppData\Local\Programs\Python\Python38-32\filess>notepad 111.txt
```

Hit on enter then it shows the following whether to open or not?



Click on “yes” to open else “no” to cancel

Write a python program to open and read a file

```
a=open(“one.txt”,”r”)
```

```
print(a.read())
```

```
a.close()
```

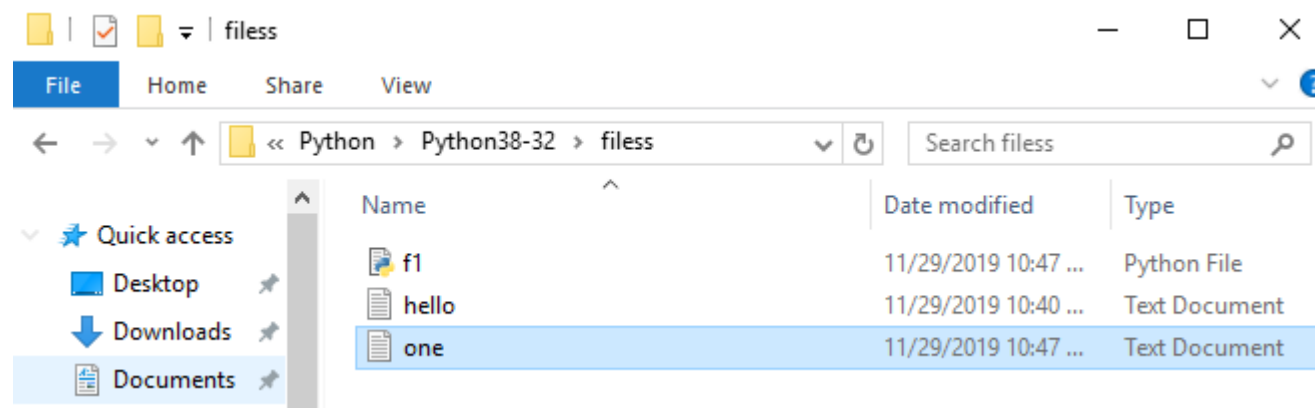
Output:

```
C:/Users/MRCET/AppData/Local/Programs/Python/Python38-32/filess/f1.py  
welcome to python programming
```

(or)

```
C:\Users\MRCET\AppData\Local\Programs\Python\Python38-32\filess>python f1.py  
welcome to python programming
```

Note: All the program files and text files need to be saved together in a particular file then only the program performs the operations in the given file mode

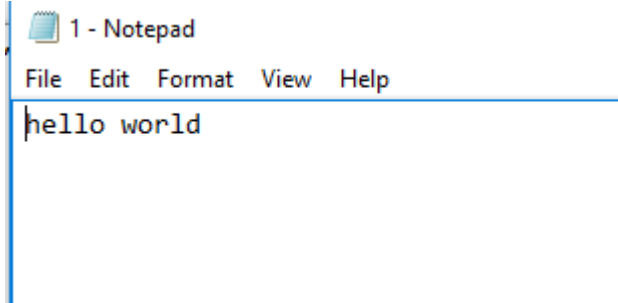


f.close() ---- This will close the instance of the file somefile.txt stored

Write a python program to open and write “hello world” into a file?

```
f=open("1.txt","a")  
f.write("hello world")  
f.close()
```

Output:



(or)

```
C:\Users\MRCET\AppData\Local\Programs\Python\Python38-32\filess>type 1.txt  
hello world
```

Note: In the above program the 1.txt file is created automatically and adds hello world into txt file

If we keep on executing the same program for more than one time then it append the data that many times

```
C:\Users\MRCET\AppData\Local\Programs\Python\Python38-32\filess>type 1.txt  
hello worldhello world
```

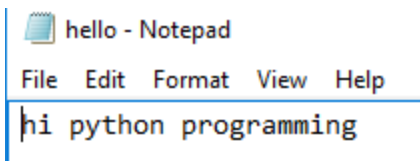
Write a python program to write the content “hi python programming” for the existing file.

```
f=open("1.txt",'w')
```

```
f.write("hi python programming")
```

```
f.close()
```

Output:



In the above program the hello.txt file consist of data like

```
C:\Users\MRCET\AppData\Local\Programs\Python\Python38-32\filess>type hello.txt  
Hello mrcet  
good morning  
how r u
```

But when we try to write some data on to the same file it overwrites and saves with the current data (check output)

```
C:\Users\MRCET\AppData\Local\Programs\Python\Python38-32\filess>type hello.txt
hi python programming
```

Write a python program to open and write the content to file and read it.

```
fo=open("abc.txt","w+")
fo.write("Python Programming")
print(fo.read())
fo.close()
```

Output:

abc - Notepad

File Edit Format View Help

Python Programming introduced for III years

(or)

```
C:\Users\MRCET\AppData\Local\Programs\Python\Python38-32\filess>type abc.txt
Python Programming introduced for III years
```

Note: It creates the abc.txt file automatically and writes the data into it

